Course Discussions

Flood Fill (easy)

Problem Statement

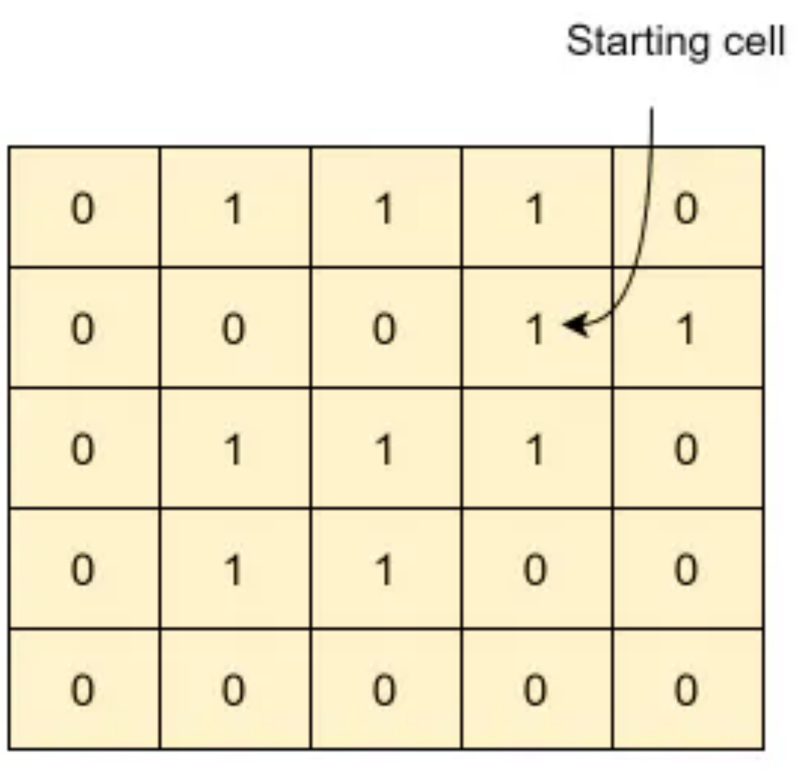
Any image can be represented by a 2D integer array (i.e., a matrix) where each cell represents the pixel value of the image.

Flood fill algorithm takes a starting cell (i.e., a pixel) and a color. The given color is applied to all horizontally and vertically connected cells with the same color as that of the starting cell. Recursively, the algorithm fills cells with the new color until it encounters a cell with a different color than the starting cell.

Given a matrix, a starting cell, and a color, flood fill the matrix.

Example 1:

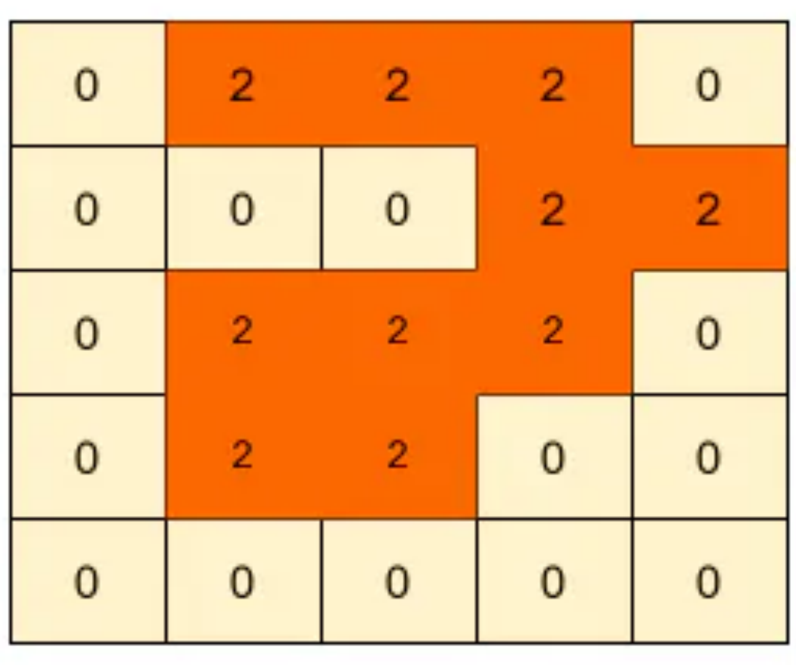
Input: matrix =



0	1	1	1	0
0	0	0	1	1
0	1	1	1	0
0	1	1	0	0
0	0	0	0	0

starting cell = (1, 3)
new color = 2

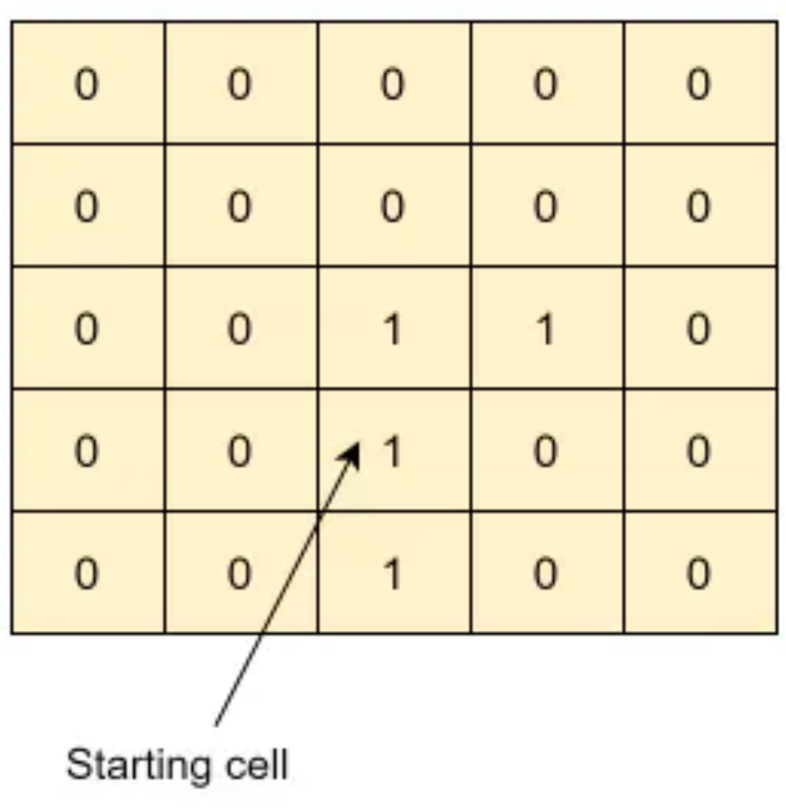
Output:



0	2	2	2	0
0	0	0	2	2
0	2	2	2	0
0	2	2	0	0
0	0	0	0	0

Example 2:

Input: matrix =

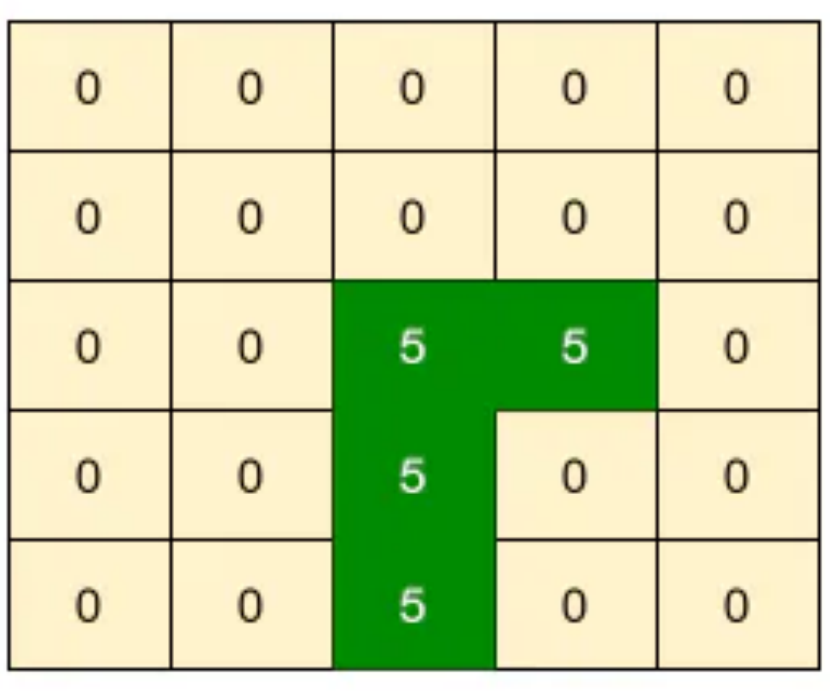


0	0	0	0	0
0	0	0	0	0
0	0	1	1	0
0	0	1	0	0
0	0	1	0	0

starting cell = (3, 2)

new color = 5

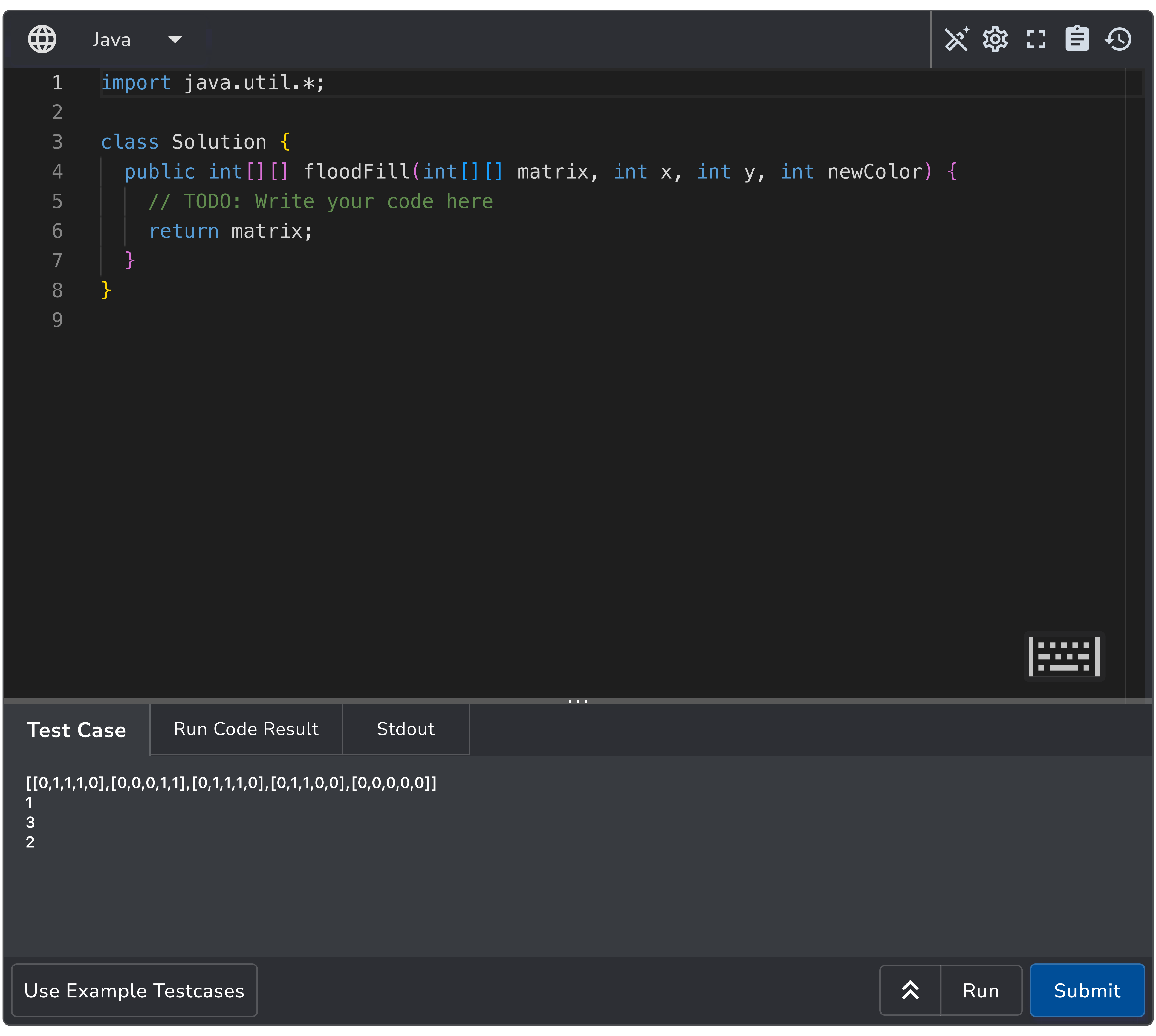
Output:



0	0	0	0	0
0	0	0	0	0
0	0	5	5	0
0	0	5	0	0
0	0	5	0	0

Try it yourself

Try solving this question here:



```
import java.util.*;

class Solution {
  public int[][] floodFill(int[][] matrix, int x, int y, int newColor) {
    // TODO: Write your code here
    return matrix;
  }
}

```

```
[[0,1,1,1,0],[0,0,0,1,1],[0,1,1,1,0],[0,1,1,0,0],[0,0,0,0,0]]
1
3
2
```

← Previous
Solution: Biggest Island

Next →
Solution: Flood Fill

☐ Mark as Completed